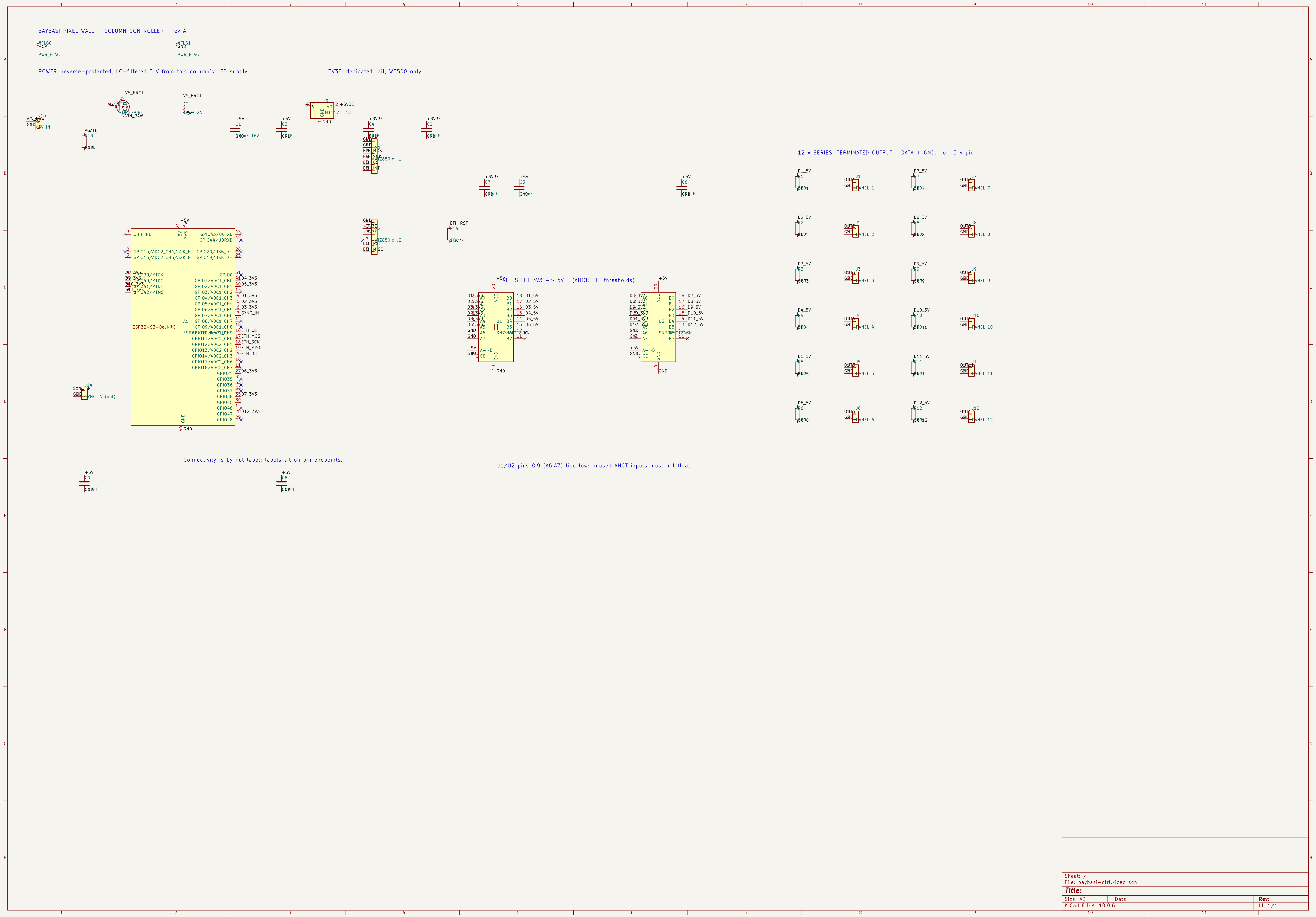
BAYBASI PIXEL WALL — COLUMN CONTROLLER   rev A
#FLG0
+5V
PWR_FLAG
#FLG1
GND
PWR_FLAG
POWER: reverse-protected, LC-filtered 5 V from this column's LED supply
3V3E: dedicated rail, W5500 only
V5_PROT
VGATE
VIN_RAW
L1
10uH 2A
J13
5V IN
R13
100k
U3
AMS1117T-3.3
+3V3E
C1
100uF 16V
C2
C3
C4
C5
C6
C7
C8
C9
100nF
10uF
W5850io J1
W5850io J2
ETH_MOSI
ETH_SCK
ETH_CS
ETH_INT
ETH_RST
ETH_MISO
R14
ESP32-S3-DevKitC
A1
CHIP_PU
GPIO43/U0TXD
GPIO44/U0RXD
GPIO15/ADC2_CH4/32K_P
GPIO16/ADC2_CH5/32K_N
GPIO20/USB_D+
GPIO19/USB_D-
GPIO39/MTCK
GPIO40/MTDO
GPIO41/MTDI
GPIO42/MTMS
GPIO0
GPIO1/ADC1_CH0
GPIO2/ADC1_CH1
GPIO3/ADC1_CH2
GPIO4/ADC1_CH3
GPIO5/ADC1_CH4
GPIO6/ADC1_CH5
GPIO7/ADC1_CH6
GPIO8/ADC1_CH7
GPIO9/ADC1_CH8
GPIO10/ADC1_CH9
GPIO11/ADC2_CH0
GPIO12/ADC2_CH1
GPIO13/ADC2_CH2
GPIO14/ADC2_CH3
GPIO17/ADC2_CH6
GPIO18/ADC2_CH7
GPIO21
GPIO35
GPIO36
GPIO37
GPIO38
GPIO45
GPIO46
GPIO47
GPIO48
SYNC_IN
J14
SYNC IN (opt)
LEVEL SHIFT 3V3 -> 5V   (AHCT: TTL thresholds)
U1
U2
SN74AHCT245N
A->B
CE
VCC
12 x SERIES-TERMINATED OUTPUT   DATA + GND, no +5 V pin
PANEL 1
PANEL 2
PANEL 3
PANEL 4
PANEL 5
PANEL 6
PANEL 7
PANEL 8
PANEL 9
PANEL 10
PANEL 11
PANEL 12
Connectivity is by net label; labels sit on pin endpoints.
U1/U2 pins 8,9 (A6,A7) tied low: unused AHCT inputs must not float.
Sheet: /
File: baybasi-ctrl.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 1/1